Table 1: **Structural comparison of benchmark protein groups** DALI metrics report Z-score, RMSD, aligned residues and sequence identity. ChimeraX MatchMaker metrics report pruned and total atom pairs, coherence (pruned/total) and global RMSD. An em dash indicates unavailable ChimeraX measurements; comparison classes are supplied by the benchmark metadata, not inferred from DALI.

Comparison	**DALI**				**ChimeraX**			
	Z	**RMSD (Å)**	**lali**	**%ID**	**Pruned**	**Total**	**Coherence (%)**	**Global RMSD (Å)**
Within: C12								
OaC12 vs MmC12	48.3	0.7	325	86	318	325	97.8	0.676
NgC12 vs OaC12	26.3	2.4	269	35	154	288	53.5	6.862
NgC12 vs MmC12	25.8	2.7	272	33	151	284	53.2	7.090
HcC12 vs Planc	22.7	1.5	180	45	155	182	85.2	3.111
Planc vs NgC12	15.5	2.7	184	30	92	200	46.0	10.147
Within: Ligase/ligase-like								
FvCOT vs 5COT	38.4	1.6	318	44	255	331	77.0	4.018
1S68 vs 5COT	18.4	2.4	184	19	78	194	40.2	8.254
5COT vs 2HVQ	17.4	2.4	185	19	76	196	38.8	8.453
5COT vs 5D1P	15.4	2.9	186	17	62	211	29.4	15.121
FvCOT vs 2HVQ	14.9	2.2	186	15	94	197	47.7	7.559
5COT vs 6N67	12.2	3.5	184	14	60	273	22.0	21.131
Between: C12 / Ligase/ligase-like								
HcC12 vs 5COT	10.0	2.7	133	17	23	168	13.7	19.631
Planc vs 5COT	8.9	2.7	134	18	49	167	29.3	14.106
HcC12 vs 2HVQ	7.2	2.8	127	17	21	145	14.5	17.606
NgC12 vs 5COT	7.1	3.3	136	18	21	243	8.6	23.215
Planc vs 2HVQ	6.9	3.1	131	16	22	145	15.2	16.619
MmC12 vs 5COT	6.7	2.8	130	18	6	176	3.4	23.158
1S68 vs OaC12	6.2	3.0	129	15	4	38	10.5	11.316
FvCOT vs OaC12	6.1	2.9	135	21	27	210	12.9	20.364
FvCOT vs NgC12	5.7	3.7	138	18	25	266	9.4	24.012
NgC12 vs 2HVQ	5.6	3.7	135	16	5	74	6.8	20.460